

CLINICAL INVESTIGATION

Eye

ALTERNATIVE DOSE FOR CHOROIDAL MELANOMA TREATED WITH AN IODINE-125 RADIOACTIVE PLAQUE: A SINGLE-INSTITUTION RETROSPECTIVE STUDY

PAUL A. SACONN, M.D.,* CHRISTOPHER J. GEE, M.D.,† CRAIG M. GREVEN, M.D.,†
THOMAS P. MCCOY, M.S.,‡ KENNETH E. EKSTRAND, PH.D.,* AND KATHRYN M. GREVEN, M.D.*

Departments of *Radiation Oncology, †Ophthalmology, and ‡Biostatistical Sciences—Division of Public Health Sciences, Wake Forest University School of Medicine, Winston-Salem, NC

Purpose: The Collaborative Ocular Melanoma Study (COMS) established iodine-125 plaque brachytherapy as an accepted standard treatment for medium-size choroidal melanoma. In the COMS, the prescription dose was 85 Gy. This is a retrospective review of our outcomes in patients treated with lower doses than those used in the COMS. **Methods and Materials:** From 1990 to 2004, 62 patients were treated with iodine-125 plaque brachytherapy for choroidal melanoma. COMS eye plaques were used with dose prescribed to the apex of the tumor. The median and average dose rates at the tumor apex were 63.5 cGy/h and 62.7 cGy/h, respectively. The median and average total doses were 63.0 Gy and 62.5 Gy (range, 56–69 Gy), respectively. The median and mean durations of implant were 100.0 hours and 101.1 hours (range, 71–165 hours).

Results: Median follow-up time was 58.2 months. The 5-year outcomes including overall survival, disease-free survival, cause-specific survival, local failure, secondary enucleation rate, and visual acuity (VA) <20/200 were estimated using the Kaplan-Meier method. Overall, there were 7 local failures, 4 distant failures, and 10 secondary enucleations (6 due to local failure and 4 due to treatment complications). Univariate analysis was performed to identify significant prognostic factors associated with disease-free survival (baseline VA in tumor eye, tumor shape), cause-specific survival (diabetic retinopathy), local failure (none found), secondary enucleation rate (diabetic retinopathy, basal tumor dimension) and VA <20/200 (diabetic retinopathy, tumor shape, age, retinal detachment, treatment depth, and history of vision-limiting condition).

Conclusions: Our survival and local control outcomes are comparable to those of the COMS. However, VA at 5 years seems to be better. Lower doses of radiation could potentially lead to better visual outcomes. © 2010 Elsevier Inc.

Choroidal melanoma, Plaque, Brachytherapy, Dose.

INTRODUCTION

Choroidal melanoma is the most common primary malignant intraocular tumor in adults. Until the 1980s, the standard treatment of choroidal melanoma was enucleation. In 1986 the Collaborative Ocular Melanoma Study (COMS) initiated a large randomized multicenter clinical trial comparing enucleation and iodine plaque brachytherapy and demonstrated equivalent survival while offering the potential for visual preservation (1). Consequently, iodine-125 plaque brachytherapy is now widely accepted as a preferred alternative to enucleation in the treatment of medium-size choroidal melanoma.

From 1986 to 1996, the COMS used a prescription dose of 100 Gy delivered to the tumor apex or 5-mm point depending on tumor height. When the Task Group (TG 43) calculation protocol became the recommendation of the American Association of Physicists in Medicine, the dose of 85 Gy was chosen

so that the actual doses delivered before and after TG 43 would be the same (2). From 1990 through 2004, we treated all patients with medium-size choroidal melanoma at our institution with a dose-reduced COMS design. This article reports the all-cause mortality, enucleation rate, and visual outcomes in our patients by the use of dosage reduction.

METHODS AND MATERIALS

A retrospective chart review of all patients with medium-size choroidal melanoma treated with iodine-125 plaque brachytherapy at Wake Forest University Baptist Medical Center from 1990 to 2004 was performed. This retrospective review was approved by the Institutional Review Board. A total of 62 patients were treated during the time period. All patients had unilateral choroidal melanoma measuring 2.0–9.6 mm in height and less than 17.0 mm in diameter. All patients were over 21 years of age and had visual acuity (VA) in the fellow eye of better than 20/200.

Reprint requests to: Kathryn M. Greven, M.D., Department of Radiation Oncology, Wake Forest University School of Medicine, Medical Center Boulevard, Winston-Salem, NC 27157. Tel: (336) 713-3600; Fax: (336) 713-6565; E-mail: kgreven@wfubmc.edu

Conflict of interest: none.

Received April 22, 2009, and in revised form Aug 20, 2009.
Accepted for publication Aug 25, 2009.